

RIJUSMIT BISWAS

New Delhi, India | rijusmit.biswas@gmail.com | +91 8929693930 | linkedin.com/in/rijusmit-biswas | github.com/riju-talk | rijusmit.vercel.app

SUMMARY

Final-year CS undergrad (B.Tech, Sept 2026) working on machine learning under real deployment constraints. I build multi-channel satellite imagery pipelines for crop classification and yield estimation, and measure how far those models degrade when capture conditions change across districts — separating real signal from capture artifacts. I train models under 4-bit quantization inside memory envelopes that match low-end hardware, and build the engineering around the model: evaluation harnesses, calibration, FastAPI inference services, and CI.

EDUCATION

B.Tech, Computer Science & Engineering, IIIT-Delhi | Coursework: Linear Algebra, Probability & Statistics, Optimisation, ML **Nov 2022 to Sept 2026**

EXPERIENCE

Satellite Imagery for Agricultural Prediction — ML (Research) Intern, CSS Lab (Econometrics), IIIT-Delhi **Jan 2025 to May 2025**

- **Pipeline:** built the full computer vision pipeline on Google Earth Engine — multi-threaded preprocessing, segmentation mask generation, and fusion with spectral bands to produce multi-channel inputs.
- **Benchmarking:** compared CNN backbones across three downstream tasks from the same imagery — phenology date prediction, crop type classification, and yield estimation — under one shared evaluation harness to keep the comparison fair. Best configuration reached `[[metric]]` on `[[task]]`, against a `[[baseline]]` baseline.
- **Generalization:** quantified performance degradation under varying capture conditions, measuring domain shift across 10+ districts with difference-in-differences and fixed-effects models to separate real signal from capture artifacts. Accuracy dropped `[[X]]` points from in-district to out-of-district evaluation, with `[[failure mode]]` as the main driver.

Long-Context Evaluation & Preference Fine-Tuning — ML Engineer / UG Researcher, MIDAS Lab, IIIT-Delhi **May 2025 to Jan 2026**

- **Training:** built training pipelines across three datasets producing 24 checkpoints with LoRA and QLoRA; merged adapters into base weights and published the artifacts for inference.
- **Constrained training:** trained Qwen3-4B under BitsAndBytes 4-bit quantization within `[[X]]` GB of GPU memory — the same envelope that governs memory-constrained and edge deployment.
- **Evaluation:** designed evaluation harnesses for long-context benchmarking; fine-tuned with DPO and GRPO and measured behavioral change against held-out baselines. `[[DPO/GRPO]]` improved `[[metric]]` by `[[X]]` while `[[holding/regressing]]` `[[other metric]]`. First-author paper under review at AAAI 2027; harness and benchmarks open-sourced.

Noisy Real-World Multi-Class Classification — Data Science Intern, Complex Systems Lab, IIIT-Delhi **Aug 2024 to Dec 2024**

- Led a team of five building a 25-class classifier on noisy real-world data, reaching 84.5% and 74.5% accuracy on two splits; traced the 10-point gap to `[[distribution shift / label noise / class imbalance]]` between splits. Ran code review and mentored junior members.

Software Engineering Intern, Shipzee **March 2026 to Aug 2026**

- Wrote unit and integration tests covering internal services and third-party APIs; diagnosed live production issues by reading logs and tracing requests across services.

PROJECTS

TURBO — Target-Augmented Shared Fusion for Multimodal Generation | PyTorch, ViT, BART, HuggingFace Transformers

- Independently reimplemented Goel et al. (arXiv:2502.07391): cross attention fusion between Vision Transformer image features and a modified BART text encoder, trained on the MORE+ dataset.
- Owned the full pipeline — feature extraction, target-conditioned representations, teacher-forced training, and BLEU / ROUGE / METEOR evaluation. Reproduced reported BLEU within `[[X]]` points; qualitative error analysis showed failures concentrated in `[[failure mode]]`.

MineMEETS — Multimodal Embedding & Retrieval Pipeline | CLIP ViT-B/32, Whisper, Pinecone, Docker, GitHub Actions, Pytest

- Built ingestion pipelines generating CLIP ViT-B/32 visual embeddings alongside text and audio behind one unified interface, with 512-dimension validation before upsert and deterministic, idempotent reprocessing.
- Set up the Pinecone vector store with namespace isolation and metadata filtering for similarity search, enabling per-item reindexing and safe rollback.
- Dockerized the service and added GitHub Actions CI running Pytest, Black, Pylint, and MyPy on every pull request; median query latency `[[X]]` ms across `[[N]]` indexed items.

PUBLICATIONS

- **HorusLLM: A Cross-Evaluation Protocol for Tri-Vector Alignment Interference.** R. Biswas (first author), A. G. Kumar, A. Anand, E. Cambria, Z. Wang, R. R. Shah. *Under review, AAAI 2027*. Built the evaluation harness and benchmarks; open-sourced.

TECHNICAL SKILLS

Computer Vision & Geospatial: PyTorch, OpenCV, CNN architectures, segmentation masks, multi-band spectral fusion, Rasterio, GeoPandas, Google Earth Engine
Deep Learning & Efficiency: HuggingFace Transformers, PEFT (LoRA / QLoRA), TRL, Accelerate, BitsAndBytes 4-bit quantization, cross attention fusion, custom training loops
Evaluation & Causal Inference: F1, ROC-AUC and PR-AUC, probability calibration, leakage-aware experiment design, SHAP, automated evaluation harnesses, difference-in-differences, fixed effects
Engineering: Python, C++, SQL (PostgreSQL), FastAPI model serving, Docker, CI/CD (GitHub Actions), Linux, PySpark, ETL pipelines

LEADERSHIP & ADDITIONAL

- Head of Product Development, IIIT-Delhi Student Council (8 engineers); Vice Chair, IEEE Student Branch. ML & DL Specializations (Stanford / DeepLearning.AI). Codeforces Specialist (1403), 2500+ LeetCode.